

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech Branch : Electrical Engineering Semester : III Subject Code & Name: BTEEC303 Electrical & Electronics Measurement Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	State and explain the types of errors	L-2/CO 1	6
B)	Discuss importance of Calibration in Measurement.	L-2/CO 1	6
C)	Describe the following terms. (i) Precision (ii) Drift (iii) Resolution	L-2/CO 1	6
Q.2	Solve Any Two of the following.		12
A)	Describe the working principle of single phase induction type energy meter.	L-2/CO 2	6
B)	What is difference between PMMC & MI Instruments?	L-2/CO 2	6
C)	Write short note on CT and PT	L-2/CO 2	6
Q. 3	Solve Any Two of the following.		12
A)	With neat sketch explain the construction and working of megger	L-2/CO 2	6
B)	Explain the method of measuring medium resistance with neat diagram and equation of Wheatstone bridge at balance.	L-2/CO 2	6
C)	Explain the method of measuring low resistance with neat diagram and equation of Kelvin's Double bridge at balance.	L-2/CO 2	6
Q.4	Solve Any Two of the following.		12
A)	Draw a block diagram of CRO and Explain the function of each block.	L-2/CO 2	6
B)	What is difference between Digital & Analog Instruments ?	L-2/CO 2	6
C)	Draw a block diagram of DVM and Explain the function of each block.	L-2/CO 2	6
Q. 5	Solve Any Two of the following.		12
A)	Explain the selection criteria for the transducers	L-2/CO 3	6
B)	Write short note on LVDT.	L-2/CO 3	6
C)	Explain working of Strip Chart Recorder.	L-2/CO 3	6